CanonGate — System Architecture Blueprint

Autonomous Pre-Publication Brand & Regulatory Claim Governance Platform • Production Technical Reference

Version: 2.0 • Standard: RFC 9162 / MCP 2026-07-28 • License: MIT
Core Stack: Python 3.12 / FastAPI / Z3 SMT / React 18 / Conformal Risk Engine

1. Multimodal AST Layer — Deterministic Parsing

Omnichannel Ingestion Adapters

- Plaintext & Copy: Markdown, TXT, Raw
- Structured Web: HTML5, DOM walker
- Office Docs: DOCX (`python-docx`), PPTX
- Financial/PDF: PDF (`pypdf` + `pdfplumber`)
- Tabular: CSV, TSV, Markdown tables

Visual Computer Vision (OpenCV + OCR)

- Contrast Analysis: WCAG 2.1 AA (4.5:1 / 3:1)
- Font Height Metric: Bounding-box ratio
- Color Palette: CIEDE2000 ΔE delta
- Text Recovery: Tesseract OCR engine
- Visual Placement: Banner fine print audit

Normalized ContentAST

- Frozen Immutable AST: Zero side-effects
- Node Tree: Text, Heading, Image, Table
- Locator Tracking: Exact char/line spans
- Deterministic Hash: SHA-256 content IDs

2. Canon Governance Domain — Signed YAML

Brand & Legal Canons (YAML)

- Digital Lending: RBI Fair Practices Code
- D2C Nutra/Pharma: ASCI / FDA claim norms
- Enterprise Core: Brand voice, design tokens
- Cryptographic Seal: Ed25519 Canon signature
- Dynamic Compilation: Pipeline A auto-gen

Multi-Constraint Rule Catalog

- Term Rules: Banned words, approved synonyms
- Typography: Minimum px height & contrast
- Palette Rules: Primary/secondary HEX bounds
- Obligations: Trigger → mandatory disclosures
- Numeric Limits: APR caps, min tenure constraints
- Fact Catalog: Grounded enterprise ground truths

Domain Boundaries (Import-Linter)

- `canongate.domain` is pure & frozen
- Zero I/O, zero network, zero internal imports
- Strict architectural boundary enforcement

3. Tri-Tier Verification Engine — Hybrid Formal + SMT

Tier 1: Deterministic Static Checks (<5ms)

- Regex & AST Walkers: Banned phrases & claims
- Visual Contrast: RGB luminance & font ratios
- Table Integrity: Header alignment & empty cells
- Trigger Matcher: Deterministic requirement trigger

Tier 2: Z3 SMT Prover & Slot NLI (<25ms)

- Z3 SMT Solver: Symbolic numeric & interval logic
- Disjunctive Satisfiability: Contradiction proofs
- Claim Slot NLI: MiniCheck entailment verification
- Sub-claim Decomposition: Multi-fact isolation

Conformal Risk Control Gate (CRC)

- Miscoverage Bound: Split-conformal $\alpha \in [0.01, 0.25]$
- Mathematically Calibrated: False-positive ceiling
- Dynamic Thresholding: $\lambda^*(\alpha)$ risk slider
- Two-Tier Verdict: Blocker ($s \geq 50$) vs Warning ($s < 50$)

Tier 3: SOTA 6-Gate Surgical Rewriter

- DPDP 2025 PII Scrubber: Local PAN/GSTIN/Aadhaar/UPI
- 6 Quality Gates: Length ratio, PII integrity, anchor lock, edit ceiling <= 35%, re-verification, zero collateral
- Gemini 3.x Cascade: 3.8-flash → 3.6-flash → 3.5-flash
- Circuit Breaker: Failover in < 1ms to deterministic AST

4. Cryptographic Trust Ledger — RFC 9162 Merkle

RFC 9162 Merkle Tree Audit Trail

- Cryptographic Accumulator: Binary tree structure
- Audit Proof: $O(\log N)$ inclusion proof path
- Consistency Proof: Append-only immutability
- Root Hash Anchor: External transparency log

SHA-256 Hash-Chained Workflow

- Deterministic Hash Chain: $H_i = \text{SHA-256}(H_{i-1} \parallel E_i)$
- Friction Log: Justification & time-penalty ledger
- State Machine: Table-driven lifecycle transitions
- Zero Tamper Window: Offline verifiable certificate

C2PA & Export Credentials

- Content Credentials: Signed pre-publish manifest
- Offline Verification: CLI `canongate verify cert.json`
- Courtroom-Ready: Cryptographic audit bundle

5. Surfaces & Endpoints — Omnichannel

Pre-Publish Webhook Gate

- `POST /api/gate/check`
- Shopify Adapter: Products, variant prices, images
- Webflow Adapter: CMS fields, rich-text, hero assets
- CI/CD Gate: GitHub Actions / GitLab pipeline blocker

Model Context Protocol (MCP)

- Stdio CLI: `canongate mcp`
- HTTP Transport: `POST /api/mcp`
- 6 Tools: `verify_campaign`, `list_canons`, `inspect_canon`, `propose_fix`, `explain_rule`, `batch_evaluate`
- Compatibility: Claude Desktop, Cursor, Antigravity

Enterprise Studio UI

- React 18 + Vite: Tailwind CSS SPA
- 3 Personas: Creator, Brand Mgr, Legal Counsel
- Interactive Studio: In-situ highlights, diff rewrites, Merkle visualizer, α-slider

< 18 ms	100% Pass	O(log N)	Zero-Data-Exfiltration
Deterministic Pre-Flight Gate Latency	114/114 Backend Tests & 3/3 Import Contracts	RFC 9162 Merkle Audit Inclusion Verification	Air-Gapped DPDP 2025 Local Scrubber & Offline Execution